

LedLamp — rebuilding a Chinese ceiling lamp as a Zigbee tunable-white light

Keep the lamp's **metal case** and diffuser; replace everything inside with:

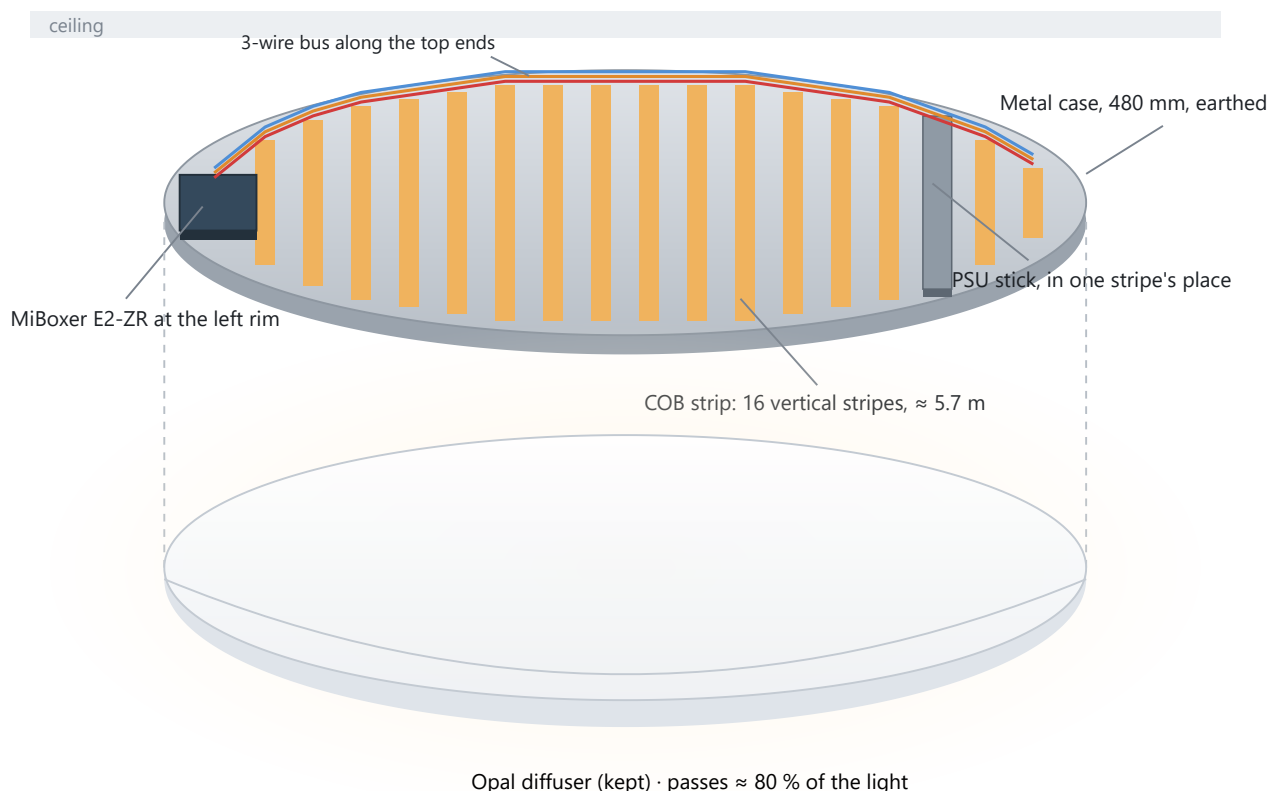
1. **COB LED strip, tunable white (CCT)**: warm 2700 K to cool 6500 K, dot-free under the diffuser.
2. **Zigbee CCT controller**: paired with the house's Zigbee2MQTT, so HA gets brightness + colour temperature.
3. **Slim 24 V power supply**.

Constraints: living room (IP20 parts are fine, nothing sealed needed), **Zigbee only**, **≤ 2500 UAH** in total, and as bright as possible within that.

The lamp: Luxel **CLRR-72**, a round remote-controlled fitting, **480 mm across** (measured) and about 50 mm deep. Its label says **72 W, 175–260 V, 5400 lm**.

What's inside now (from a photo of the opened lamp): two square LED boards, about 33 cm and 24 cm across, carrying roughly **75 separate LEDs under small lenses**, about 2.3 m of board in total. A small mains driver in a clear plastic box and a 3-way terminal block (L, N, earth) sit in the middle, and the base is painted white metal. There are mounting holes near the rim at the top, bottom and right. The separate lensed LEDs are why the old light looks spotty.

Prices and stock were checked on **2026-09-26** and change often, especially on prom.ua. Rozetka refused automated lookups (HTTP 403). A printable version with drawings, [LedLamp.pdf](#), is built from this README and `docs/` by `tools/build_pdf.py`.



1. Parts to buy: ≈ 3200 UAH for the lamp, ≈ 825 for the wall switch

#	Part	Pick	Shop	Price, UAH
1	CCT COB strip, 7 m (5.7 m used, 1.3 m spare)	LEDTech 24V COB/FCOB CCT 2700–6500K Multi White: 14 W/m (7+7), 1400 lm/m, 608 LED/m, Ra>90, 10 mm wide	prom.ua (LEDTechnics)	~1578 (225.40/m × 7 m)
2	Zigbee controller	MiBoxer (Mi-Light) E2-ZR W+CCT 2-in-1: Zigbee 3.0 + 2.4G RF, DC 12–24 V, 12 A, 100.6×40×17.6 mm , PWM 250 Hz / 16 kHz	prom.ua (OPTSVET)	1210
3	Power supply	BIOM Professional STICK BPBLS-60-24: 24 V / 2.5 A, 313×15×18 mm , IP20, metal case, 176–265 V in	samsnab (order via Viber) · biom.ua, 358	315
4	Wire (0.75 + 0.5 mm ²), heat-shrink, 3 Wago 221	local	—	~100
Total (lamp)				≈ 3203

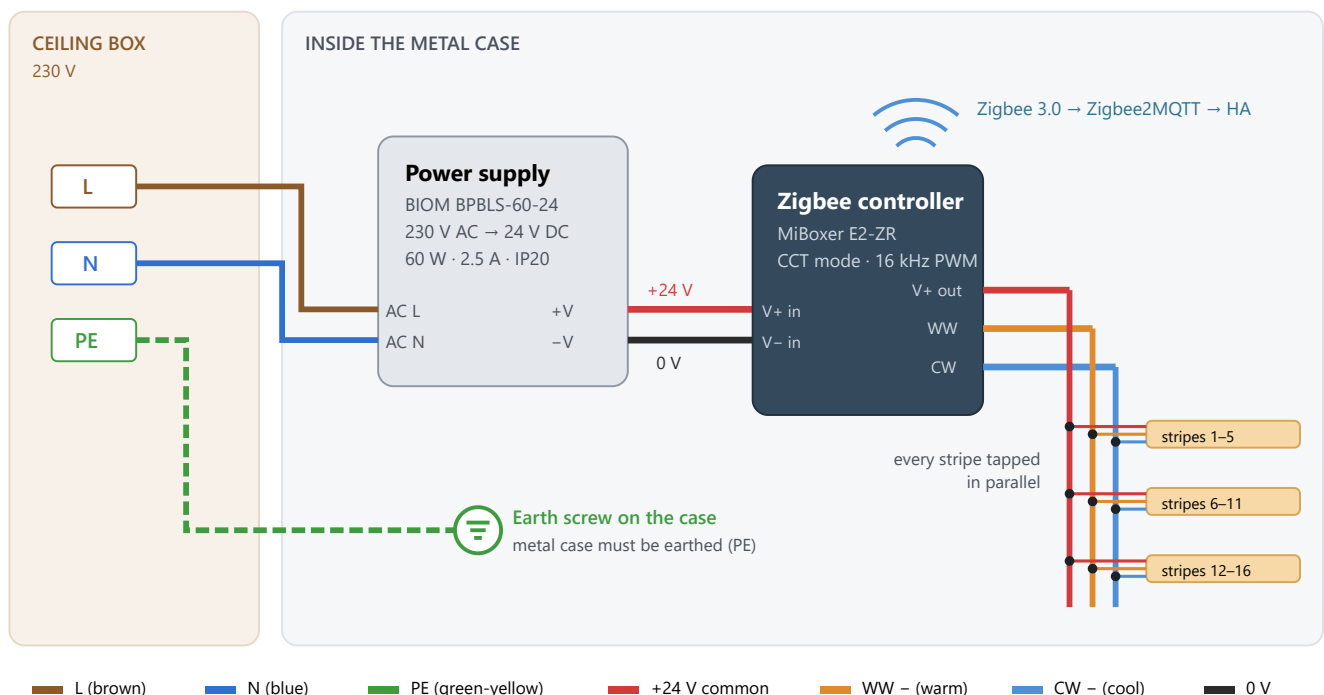
Ask the strip seller for 7 m. The layout uses about 5.7 m; the rest covers cutting mistakes and a spare stripe for repairs. The listing is priced per metre; if they only sell whole 5 m reels, two reels (10 m) cost ~2254 and the total rises to ≈ 3880. The spare strip, the better power supply and the E2-ZR put the lamp about 700 over the original 2500 budget; the wall switch ([docs/wall-switch.md](#)) adds ≈ 825. The [LT COB-24-MW-608](#) at [svetum](#) (same 7+7 W/m, CRI 90, 10 mm) is sold only in 5 m steps.

Why these: - **Strip:** the only one found that meets every point (2700–6500 K, Ra>90, 10 mm, 14 W/m) at the lowest price. Light per metre decides the brightness, so stay at 14 W/m: 10 W/m strips give about a third less. - **Controller:** a survey of Zigbee CCT controllers found nothing meaningfully better. The E2-ZR is purpose-built for tunable white, has the highest PWM found (**16 kHz**, no flicker on camera), pairs directly with MiBoxer 2.4G RF remotes (they work even if HA is down), works as a Zigbee router, and is in stock in Ukraine. Zigbee2MQTT supports it ([E2-ZR](#): brightness, color_temp 153–500 mired, `do_not_disturb`); some units pair as the generic `TS0502B` with the same controls. No OTA updates through Z2M. - **Power supply:** the only mains part, sitting in a closed metal lamp, so it is the one worth paying a little more for. BIOM is an established Ukrainian LED brand with a local warranty, and the stick is thin enough (15 mm) to sit between two stripes. 5.7 m of strip draws ~40 W (1.7 A), so the 60 W unit runs at about two-thirds load. It is out of stock at most shops: [samsnab](#) takes orders by Viber inquiry, and [biom.ua](#) lists it at 358.

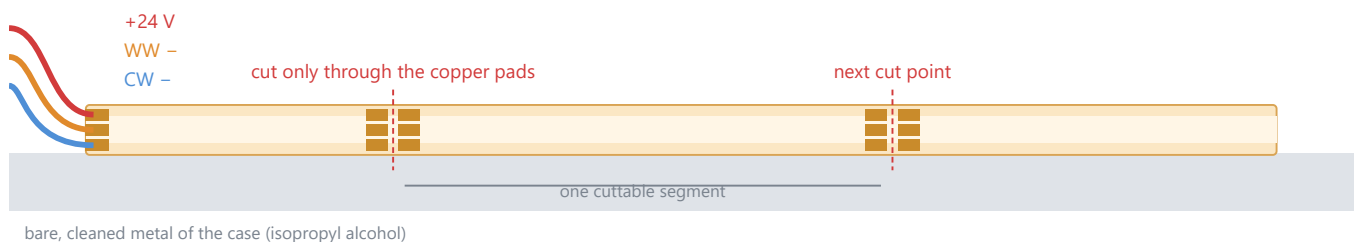
Swaps if something is out of stock: - Power supply: [LED STORY Profi 60W slim](#), 297×17×17 mm, IP20, 218 (on sale), in stock. A shop house brand with no efficiency, ripple or safety figures, so only if the BIOM can't be had. - Controller: the **Moes ZLD-RCW** ([SELLBOT, 849, Z2M](#)), a Tuya RGB+CCT unit with R, G and B left unused. Cheaper (lamp ≈ 2842), but its PWM is unpublished, batches vary, and it has no direct remote.

2. How it fits together

Wiring



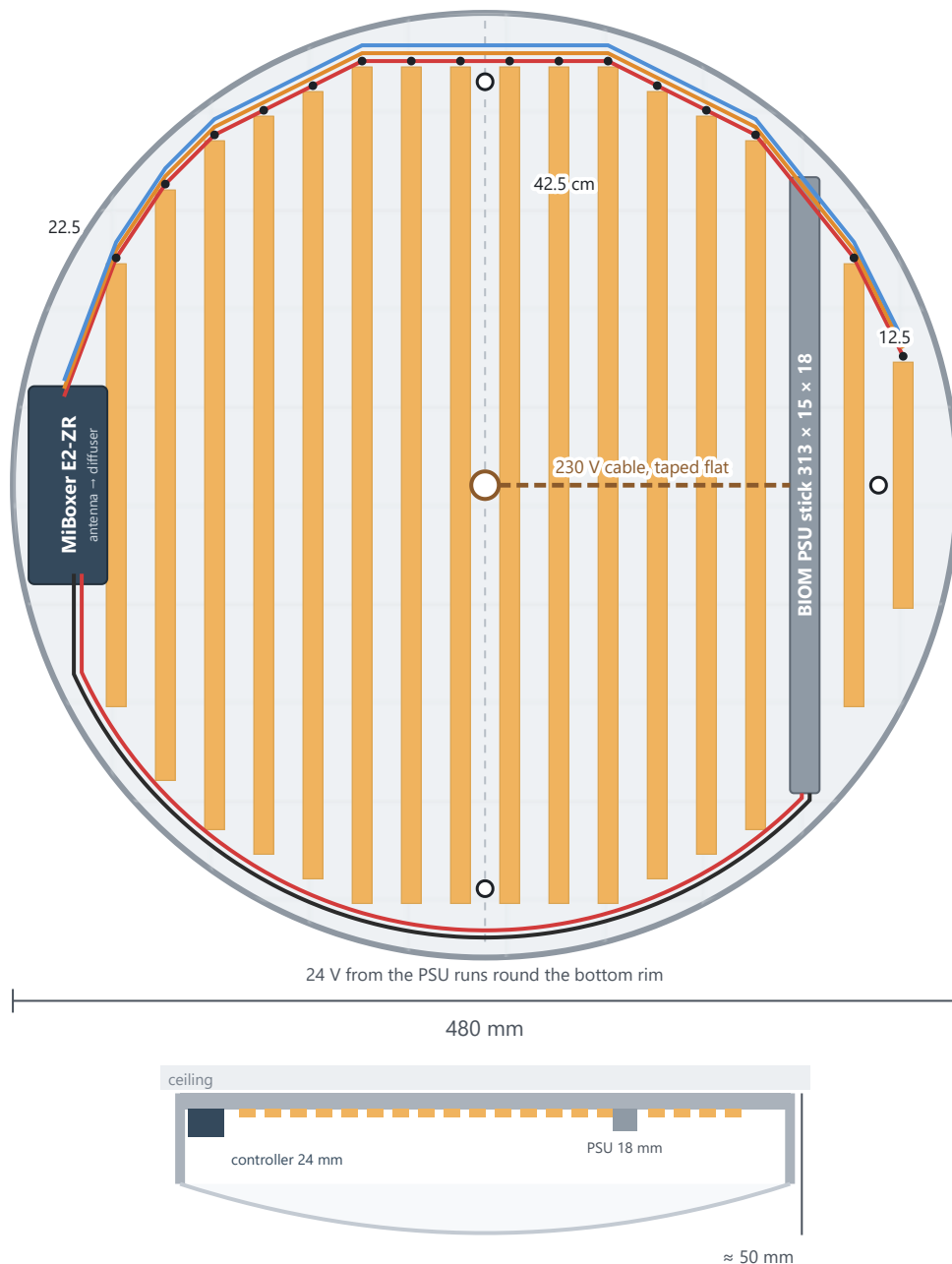
- **The strip is common-anode:** one +24 V pad and two negative pads (WW, CW). The controller switches the negatives. Follow the controller's label for which outputs are V+ and the two whites, and set it to CCT (dual white) mode.
- **Earth the case.** The PSU sits on mains inside a metal case; connect PE to the case's earth screw. The 24 V side is then touch-safe.
- **Wire gauge:** mains and PSU → controller 0.75 mm²; the 3-wire bus 0.5–0.75 mm² (1.7 A in total); tails to each stripe 0.5 mm² / 20–22 AWG.
- **Solder the strip.** Clip-on connectors are unreliable on COB strip; tin the pads and solder the three wires.



The pad order (+, WW, CW) varies between strips: read the print next to the pads, not this drawing.
Tin the pads, solder 0.5 mm² tails, cover with heat-shrink or a drop of neutral silicone.

Layout in the 480 mm case: 16 vertical stripes

bus: +24 V, WW, CW along the top ends



Strip faces down onto the diffuser. COB needs only 15–20 mm of distance to look dot-free. Height not to scale.

- **16 vertical stripes, 10 mm wide with 15 mm gaps** (25 mm centre to centre), ending about 15 mm from the wall. The full pattern is 18 stripes; the PSU stick takes one stripe's place (at $x = +16$ cm, where the case is still 339 mm tall) and the controller takes the outermost stripe on the left. The E2-ZR (100.6×40 mm) is a tight fit there: about 2–3 mm to the wall and 0.5 mm to the next stripe on paper. If your case is smaller inside, drop the 22.5 cm stripe next to it (≈ 5.45 m in total). Each stripe is cut at the nearest pads, so the lengths are 42.5 cm in the middle down to 12.5 cm at the right edge: **≈ 5.7 m in total**.
- **Keep the centre line clear:** the stripes sit at ± 12.5 , ± 37.5 ... mm, so a 15 mm gap runs down the middle. The top and bottom mounting holes fall in that gap, and the right-hand hole falls in the gap between the last two stripes. Check the holes on your case before sticking anything.
- **Feed each stripe from its top end:** a 3-wire bus (+, WW, CW) runs from the controller up the left rim and along the stripes' top ends, with a short tail down to each one. The longest stripe is 42.5 cm, so one feed per stripe keeps the brightness even. Do not daisy-chain all 16 end to end.

- **The old terminal block has to go:** it doesn't fit between two stripes. The mains cable comes through the centre hole; tape it flat across the stripes to the PSU and join it with Wago 221 connectors there.
- **Heat is not a problem.** A Zigbee CCT controller keeps warm + cool at 100 % in total, so each stripe draws at most 7 W/m: about 40 W over the whole base, ~220 W/m². Expect about 40–50 °C on the strip in a 25 °C room (an estimate; the limit is 75 °C). Strip makers only ask for aluminium above ~10 W/m, so stick it straight onto the cleaned, painted steel. A 15 mm gap is plenty; at 7 W/m even 10 mm would be safe, but narrower gaps don't add light because the length is capped by the case.
- **Zigbee through metal is weak.** Mount the controller with the antenna end toward the diffuser. It is mains-powered, so it also works as a **Zigbee router**. Its PWM frequency (250 Hz or 16 kHz) is switched with a MiBoxer RF remote; 16 kHz rules out flicker on phone cameras.
- COB strip needs little distance to the diffuser: 15–20 mm already looks dot-free.

3. Build checklist

1. **Before the rebuild**, take lux-meter readings under the old lamp at warm, neutral and cool (see [docs/brightness.md](#)).
2. Measure the free height inside and how the diffuser attaches.
3. Remove the two old LED boards, the driver and the centre terminal block. Clean the painted base where the strip will go (isopropyl alcohol).
4. Mark the centre line, then stripe centres every 25 mm from $x = \pm 12.5$ mm outward, the mounting holes, the PSU slot at $x \approx +16$ cm and the controller slot at the left rim.
5. Cut each stripe at the nearest pads so it ends ~15 mm from the wall, stick them on, and insulate every solder joint (heat-shrink or Kapton tape) so no pad touches the metal.
6. Mount the PSU and controller in their slots (thermal tape or screws). Run the 3-wire bus up the left rim and along the stripes' top ends, and solder a tail to each stripe. Earth the case, and join the mains cable at the PSU with Wago 221.
7. Bench test first on 24 V: both channels light, correct warm/cool order (swap WW/CW if reversed). At full brightness at neutral white the 24 V side should draw about 1.7 A.
8. Pair with Zigbee2MQTT (permit join, then follow the device's pairing method on its z2m page). Check that `brightness` and `color_temp` both work from HA. Set **do_not_disturb on**, so the lamp stays off when power returns after an outage instead of lighting up in the night. If you have a MiBoxer RF remote, switch PWM to 16 kHz.
9. Close the diffuser, run at full power for an hour, and check the strip is comfortably warm, not hot (under ~60 °C).
10. Repeat the lux-meter readings and compare.
11. Fit the wall switch and its automation ([docs/wall-switch.md](#)).

Tools

Script	Does
<code>tools/build_pdf.py</code>	builds <code>LedLamp.pdf</code> from this README and <code>docs/*.md</code> with headless Chrome (needs <code>pip install -r tools/requirements.txt</code>)

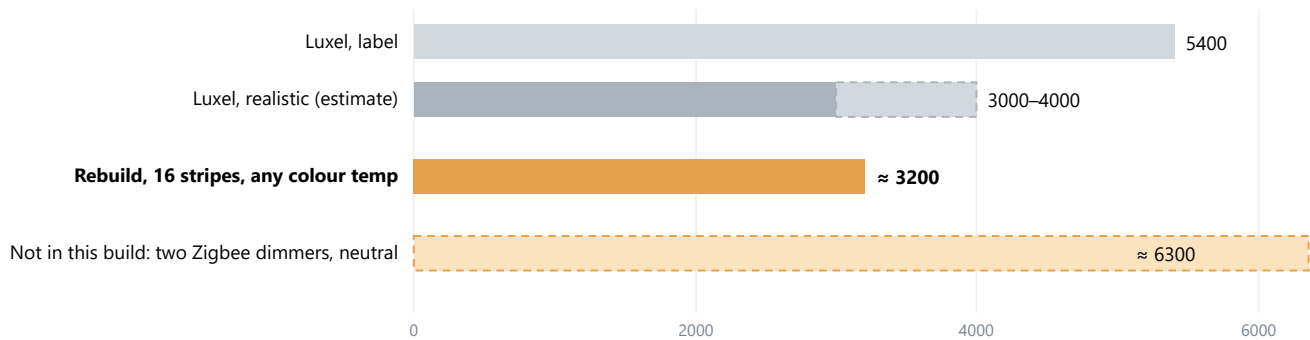
Docs

Doc	About
docs/brightness.md	how bright the rebuild will be, against the old lamp, and the numbers behind it
docs/parts-candidates.md	every strip, controller and power supply found, with prices and links
docs/option-b-mains-driver.md	the fewer-parts option (a mains Zigbee driver) and why it was not chosen
docs/wall-switch.md	the wireless rocker switch fixed into the old wall box, and its automation

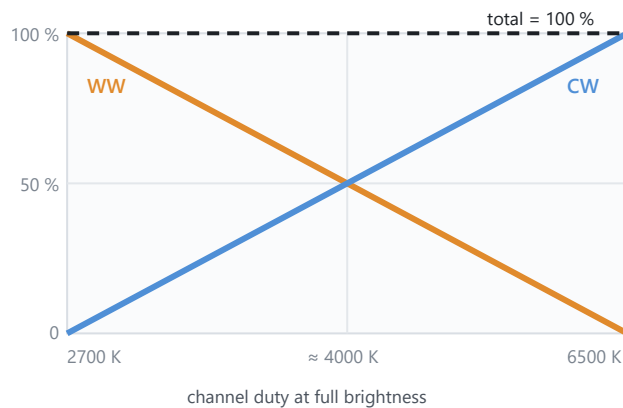
How bright will it be?

About 3200 lm out of the diffuser at any colour temperature, from about 48 W at the wall.

Light out of the diffuser, lm



How the numbers are worked out:



- **Strip:** 1400 lm/m at 14 W/m, but that is **both** channels at 100 %. A Zigbee colour-temperature controller mixes the two whites so they add up to 100 %: at 2700 K only WW is on, at 6500 K only CW, in between both share. So the usable maximum is **one channel's worth: 700 lm/m at 7 W/m**. No Zigbee CCT controller found can change this: MiBoxer confirms the E2-ZR **keeps 50 % + 50 % at neutral** and "it can not be changed".
- **Diffuser:** an opal cover passes about 80 %.
- **Power supply + controller:** about 85 % × 98 % efficient.

	LED power	From the wall	Light out of the diffuser
Rebuild, 16 stripes ≈ 5.7 m, any colour temperature	40 W	48 W	≈ 3200 lm
Luxel, label	—	72 W	5400 lm
Luxel, realistic guess	—	~45–55 W	~3000–4000 lm

The Luxel's real output: a "72 W" Chinese fitting usually draws 45–55 W, and at 70–75 lm per wall-watt that is **3000–4000 lm** with both colour channels on (neutral). That is a rough estimate, not a measurement. Against it the rebuild is **about the same to 20 % dimmer at neutral**, and probably brighter at the warm and cool ends, where the Luxel likely uses only half its LEDs. For an 18–20 m² living room, ~3000 lm is comfortable general light.

The one Zigbee route to more light at neutral: two single-channel Zigbee dimmers (e.g. 2× MiBoxer E2-ZR in single-colour mode, one per white) mixed by an HA template light, so both whites can run at 100 %. That gives up to ~6300 lm at neutral from ~80 W, but needs a second E2-ZR (+1210), a 100 W supply, and at 14 W/m the strip wants aluminium under it. Not part of this build.

Measure instead of guessing: put a phone lux-meter app on the floor under the lamp, and write down the old lamp's readings at warm, neutral and cool. Repeat after the rebuild at the same spot.

All candidates found

CCT COB strip, 24 V, 3-wire

Product	Shop	UAH/m	Specs	Notes
LEDTech 24v COB/FCOB CCT 2700–6500K	prom.ua	225	14 W/m, 608 LED/m, Ra>90, 10 mm	Pick
LT COB-24-MW-608, 7+7 W/m	svetum.com.ua	216	14 W/m, CRI 90, 10 mm, IP20	Sold in 5 m steps only
RISHANG RM0821MC-L 2700/6500K	prom.ua	213	10 W/m, Ra>90, 8 mm	A third less light per metre
RISHANG RM0831MC-A-1 2700/5700K	prom.ua	181	10 W/m, 950 lm/m	Only to 5700 K
ISOBO V-PRO COB 608 CCT	prom.ua (VOLT)	275	7 W/m, CRI≥90	Half the light per metre
COLORS D8140SWW, 2835 SMD, 19.2 W/m	lightland.com.ua	311	1710 lm/m, CRI 90	Most light per metre, but 1553 per reel breaks the budget; SMD, not COB

Zigbee CCT controllers

Product	Shop	UAH	Specs	Zigbee2MQTT
Moes ZLD-RCW RGB+CCT	prom.ua (SELLBOT) · Ledenerho, 985	849	12–24 V, 6 A/ch, 90×40×24 mm	ZLD-RCW_1
MiBoxer E2-ZR W+CCT 2-in-1 (pick)	prom.ua (OPTSVET) · VITRINA, 1342	1210	12–24 V, 12 A, 17.6 mm thick, + RF	E2-ZR
Aqara Dimming Module T1 (ZNDDMK11LM)	prom.ua (SELLBOT)	1599	12/24 V, 75×33×19 mm	ZNDDMK11LM : warmest ≈ 2700 K
Gledopto GL-C-203P (= GL-C-006P)	prom.ua	780	Pro+ CCT	GL-C-006P . Out of stock
Sunricher SR-ZG9101CS (single channel)	svl.ua	1578	12–36 V, 8 A	SR-ZG9101CS ; only for the two-dimmer route

Slim 24 V power supplies, IP20

Product	Shop	UAH	Size, mm	Notes
LED STORY Profi 60W slim	led-story.ua	218	297×17×17	Fallback; house brand, in stock
BIOM STICK BPBLS-60-24	samsnab · biom.ua, 358	315	313×15×18	Pick ; known brand; Viber inquiry at samsnab, out of stock elsewhere
COXO DS-60-24	lightland	585	130×53×27	CE/CQC, 85 %, 3-year warranty; over budget
EUROLAMP PS 60W 24V 2.5A	eurolamp.ua	469	116×53×27	Out of stock; rated only to +40 °C
No-name XJK-60-24	prom.ua	206	123×34×22	Risky: ripple and inflated rating

Wall switch: classic rockers, for an elderly family member

The lamp's controller must stay powered, so the old mechanical switch goes: its two wires are joined permanently and a **battery Zigbee switch that looks like an ordinary two-key rocker switch** takes its place, **fixed into the old 60 mm box**.

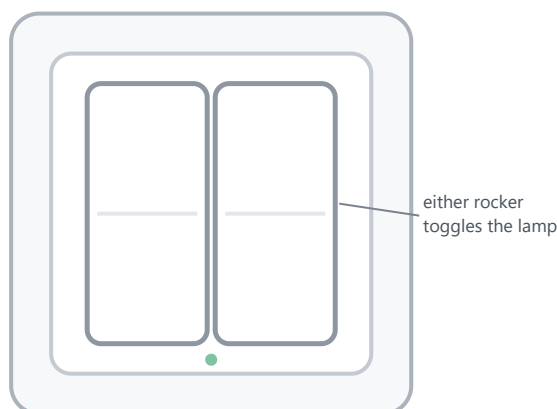
Parts: ≈ 825 UAH

Part	Pick	Shop	Price, UAH
Switch	Aqara Wireless Remote Switch E1, two rockers (WXKG17LM): two rockers on an 86×86×15 mm plate, CR2032, Zigbee 3.0	SELLBOT on prom.ua , ready to ship (single-rocker WXKG16LM: 600 at sellbot)	650
Box adapter	Schneider Asfora blank EPH5600121 , white, 83×83 mm, fixes into the box with spreading claws	schneider.kiev.ua · Epicentr · 5watt , 133	134
Small parts	3M VHB tape, spare CR2032, 1 Wago 221	local	~40
Total			≈ 824

Why these: - **Looks like the switch they know:** rockers you press, like the classic two-key switch, not a flat touch panel or a small button. For only ≈ 50 more than the single-rocker E1 it matches a two-key switch, and **both rockers do the same thing** (toggle the lamp), so whichever side gets pressed, it works. It has a "fast" click mode in which a single press acts at once instead of waiting to see if a double press follows. Z2M: [WXKG17LM](#) (single, double and hold per rocker, plus both at once). - **Why the blank plate:** Soviet-era boxes have **no screw lugs**; switches were held by spreading claws. So nothing can screw into the hole. The Asfora blank clamps into the box with its own claws, and the switch is bonded onto its flat plastic face. The blank does the bolting; the switch never relies on crumbly plaster. - **Alternatives:** the [Moes 1-button TS0041 panel](#) (690, screw holes, but a flat push button), the single-rocker Aqara E1 WXKG16LM (600), and the Aqara H1 single rocker (binds directly to the lamp, so it works without HA, but it isn't sold in Ukraine; the two-rocker H1 costs ~1299). IKEA RODRET is cheaper but has separate on/off buttons.

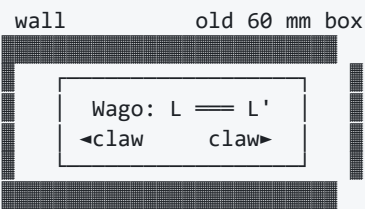
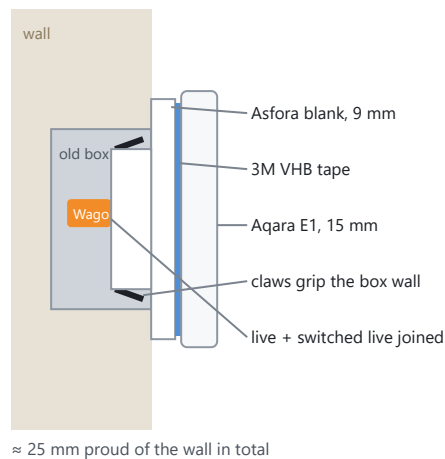
Mounting

Front



Aqara E1 two rockers (86 mm) on the Asfora blank (83 mm)

Side section



claws pushed out
against the box wall

Stack, side view:

wall		Asfora blank	VHB	Aqara E1
		9 mm		1mm 15 mm
		claws inside the box		
11	≈ 25 mm proud of the wall in total			

1. **Power off** at the breaker. Take out the old switch.
2. Join the two wires (live and switched live) permanently in a **Wago 221**, and push it to the back of the box.
3. Insert the **Asfora blank** and tighten its claw screws evenly. Old metal boxes are smooth and claws can slip; if it feels loose, add a bead of glue or wrap the claws for grip.
4. Wipe the blank's face with isopropyl alcohol, stick **3M VHB** to the back of the switch plate, and press it on level for 30 seconds. Leave it 24 hours before heavy use.
5. Power back on (the lamp is now always live), pair the switch in Zigbee2MQTT, name it `Living room switch`, and set its **click mode to fast** if Z2M offers it.

The automation (Home Assistant)

A sketch: adjust the entity and device names to yours.

```
automation:
- alias: "Living room ceiling: wall switch"
  mode: single          # debounce: presses during the delay are ignored
  max_exceeded: silent
  triggers:
  - trigger: mqtt
    topic: "zigbee2mqtt/Living room switch"
    # either rocker (or both at once) toggles the lamp
    value_template: "{{ value_json.action in ['single_left', 'single_right', 'single_both'] }}"
    payload: "True"
  actions:
  - if:
    - condition: state
      entity_id: light.living_room_ceiling
      state: "on"
    then:
    - action: light.turn_off
      target: {entity_id: light.living_room_ceiling}
    else:
    - action: light.turn_on
      target: {entity_id: light.living_room_ceiling}
      data: {brightness_pct: 100, color_temp_kelvin: 3000}
  - delay: {seconds: 1}
```

Setting it up for an elderly user

- **One action only:** a single press on **either rocker** toggles. Double press and hold do nothing. Don't give the second rocker a different job; two rockers that behave differently are confusing.
- **Always the same light:** "on" is always 100 % at 3000 K, never last night's dim setting.
- **Debounce:** presses within 1 s of the last one are ignored, so a double press by mistake doesn't switch the light straight back off.
- **Mount it exactly where the old switch was;** habit leads them there.
- **Battery:** replace the CR2032 on a schedule (e.g. every 12 months) and add an HA low-battery alert for the caregiver.
- **After a power cut the lamp stays off** (`do_not_disturb` on, see the [build checklist](#)), so it doesn't wake anyone at night.
- **The one real risk:** this switch works through HA. If HA or Zigbee2MQTT is down, pressing it does nothing. Keep HA on a UPS, and give the caregiver a second way to switch the lamp (the HA app, or a MiBoxer RF remote paired straight to the E2-ZR).

Option B, fewer parts: a mains Zigbee CCT driver

One box instead of PSU + controller: a **230 V Zigbee constant-current CCT driver** (Tuya TS0502B type, e.g. DC 72–185 V, 260 mA, 2 channels with common +) driving **high-voltage 3-pin chandelier strip**, for example the [Electro Lab 3 PIN 5B10C/5B20C](#), 85 UAH/m (3000 + 6500 K, 60 V per metre). It works in Z2M (TS0502B) and could cost about 600–900 UAH, but **no Ukrainian seller of the Zigbee driver was found** (AliExpress/Taobao only), the strip carries up to ~185 V DC on exposed pads, and cheap drivers are often not isolated from mains. Not recommended.
